Lab 10: Approval must never post before the passing test artifact exists

Topic 1: Agentic Architecture & Orchestration · Task 1.4: Implement multi-step workflows with enforcement and handoff patterns

Runs on Simulator 5: Claude Code for Continuous Integration · Run in class

Source: SkillCertPro practice exam Part 1, Q3 (unofficial third-party question)

The question

A compliance review finds that, in rare pull requests, the system posts an approving review before the required test-result artifact has been attached. The rule is non-negotiable: approval actions must not execute unless the artifact is already present and passing. What change should this team make?

A. Add a stronger system prompt that instructs Claude to verify artifacts before approvals.

B. Add a PreToolUse command hook that denies approval commands when the passing artifact is absent.

C. Add a UserPromptSubmit command hook that appends check instructions to approval requests.

D. Add a PostToolUse command hook that checks the artifact after approval commands complete.

Simulate the solution

```
# In the browser: open Simulator 5, then Lab 10
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 10

# In a terminal, from simulators/simulator-5-ci-cd/
uv run hook_gate.py
```

What you should see

No artifact: exit 2, denied. Tests run first: PR 418 approved and logged.

Scenario 1: the hook exits 2, Claude quotes 'DENIED by require-passing-tests hook' and ci/approvals.log never appears. Scenario 2: after ./ci/run_tests.sh the same prompt approves PR 418 and approvals.log gains one line.

Open `simulators/simulator-5-ci-cd/evidence/` and find it yourself.

The answer

B. A PreToolUse command hook denies approval while the artifact is absent.

A non-negotiable rule needs an enforcement point outside the model's voluntary behaviour. Claude Code hooks give deterministic control at lifecycle points. PreToolUse fires before the tool runs and can block the call: a command hook reads the event JSON on stdin, checks whether the passing artifact exists, and denies the action with feedback if it does not.

Why the others fail

A. Still relies on model compliance. “Rare” failures are exactly what probabilistic guidance leaves behind.

C. Only adds instructions to the prompt. It still depends on the model choosing correctly and never blocks the tool call itself.

D. PostToolUse fires after the tool succeeded, so the approval has already been posted.

Check yourself

- without the artifact the approval command is blocked before it runs (exit 2, reason fed to Claude)
- with a passing artifact the same command runs and records the approval
- approvals.log carries exactly the approval line
- learner explains why PostToolUse is too late and why a prompt or UserPromptSubmit hook only appends instructions

Now change one thing

Fix the loop in repo/stock/totals.py and the caller in repo/stock/jobs.py, rerun, and watch the blocking count fall.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.